

LINEAR ALGEBRA: Basics for Regression Analysis

Theorem 4

$$\text{Let } A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad \begin{aligned} A_{11 \cdot 2} &= A_{11} - A_{12} A_{22}^{-1} A_{21} \\ A_{22 \cdot 1} &= A_{22} - A_{21} A_{11}^{-1} A_{12} \end{aligned}$$

$$A^{-1} = \begin{bmatrix} A_{11 \cdot 2}^{-1} & -A_{11 \cdot 2}^{-1} A_{12} A_{22}^{-1} \\ -A_{22 \cdot 1}^{-1} A_{21} A_{11}^{-1} & A_{22 \cdot 1}^{-1} A_{12} A_{22}^{-1} \end{bmatrix} \quad (*)$$

$$= \begin{bmatrix} A_{11}^{-1} + A_{11}^{-1} A_{12} A_{22 \cdot 1}^{-1} A_{21} A_{11}^{-1} & -A_{11}^{-1} A_{12} A_{22 \cdot 1}^{-1} \\ -A_{22 \cdot 1}^{-1} A_{21} A_{11}^{-1} & A_{22 \cdot 1}^{-1} \end{bmatrix}$$

(proof)

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \underbrace{\begin{bmatrix} I & 0 \\ A_{21} A_{11}^{-1} & I \end{bmatrix}}_L \underbrace{\begin{bmatrix} A_{11} & 0 \\ 0 & A_{22 \cdot 1} \end{bmatrix}}_D \underbrace{\begin{bmatrix} I & A_{11}^{-1} A_{12} \\ 0 & I \end{bmatrix}}_U$$

* LDU decomposition of matrix A

$$\therefore A^{-1} = (L D U)^{-1} = U^{-1} D^{-1} L^{-1} = \begin{bmatrix} I & -A_{11}^{-1} A_{12} \\ 0 & I \end{bmatrix} \begin{bmatrix} A_{11}^{-1} & 0 \\ 0 & A_{22 \cdot 1}^{-1} \end{bmatrix} \begin{bmatrix} I & 0 \\ -A_{21} A_{11}^{-1} & I \end{bmatrix}$$

$$\text{How to find } L^{-1}. \text{ Suppose that } L^{-1} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

$$L \cdot L^{-1} = I$$

$$\therefore \begin{bmatrix} I_{11} & 0 \\ A_{21} A_{11}^{-1} & I_{22} \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} I_{11} A & I_{11} B \\ A_{21} A_{11}^{-1} A + I_{22} C & A_{21} A_{11}^{-1} B + I_{22} D \end{bmatrix}$$

$$= \begin{bmatrix} I_{11} & 0 \\ 0 & I_{22} \end{bmatrix} \quad \text{Solve the simultaneous equation.}$$

$$A = I_{11}, B = 0, C = -A_{21} A_{11}^{-1}, D = I_{22}$$

$$\therefore L^{-1} = \begin{bmatrix} I & 0 \\ -A_{21} A_{11}^{-1} & I \end{bmatrix}$$

* UDL decomposition of matrix A in the precedent derivation of the inverse matrix would give (*)

Theorem 5 (Sherman - Morrison Lemma)

Let $a, b \in \mathbb{R}^n$ be vectors and $C \in \mathbb{R}^{n \times n}$ be an invertible matrix.

$$(I_n + ab^T)^{-1} = I_n - \frac{ab^T}{1 + b^T a}$$

$$(C + ab^T)^{-1} = C^{-1} - \frac{C^{-1} ab^T C^{-1}}{1 + b^T C^{-1} a}$$

(proof)

$$\begin{aligned} (I + ab^T) \cdot (I - \frac{ab^T}{1 + b^T a}) &= I - \frac{ab^T}{1 + b^T a} + ab^T - \frac{ab^T ab^T}{1 + b^T a} \\ &= I - \frac{ab^T}{1 + b^T a} + ab^T - b^T a \frac{ab^T}{1 + b^T a} \\ &= I + (1 - \frac{1}{1 + b^T a}) ab^T - (\frac{b^T a}{1 + b^T a}) ab^T \\ &= I + (\frac{b^T a}{1 + b^T a}) ab^T - (\frac{b^T a}{1 + b^T a}) ab^T = I \end{aligned}$$

$$(C + ab^T)^{-1} = (I + a' b'^T)^{-1} C^{-1}, \quad a' = C^{-1} a$$

$$= (I - \frac{a' b'^T}{1 + b'^T a'}) C^{-1} = (C^{-1} - \frac{C^{-1} ab^T C^{-1}}{1 + b^T C^{-1} a})$$